

Advanced Excel Project Report

Mapple Employee Performance & Payroll Management Analytics

Student Name	Somnath Mondal, Arpita Dhar	AF Code	AF05325748, AF05325379
Project Title	Employee Performance & Payroll Dashboard	Project Duration	4 Days
Trainer Name	Sayan Sau	Submission Date	28/08/2026
GitHub Username	Somnaths13, adarpitadhar13-DA	GitHub Repo Link	Somnath's Repo / Arpita's Repo

1. Project Objective

This analytical project aims to construct a data-driven Excel analytics framework to evaluate work-force productivity, compensation structure, demographic distribution, and regional payroll allocations. Utilizing raw data from 2,000 employee records sourced from Kaggle, the model evaluates compensation consistency across functional departments (HR, IT, Sales), tracks experience-to-salary progression, measures appraisal coverage gaps, and models employee retention dynamics across major corporate locations.

2. Problem Statement

Modern enterprise HR operations frequently face structural inefficiencies due to disjointed compensation models, missing appraisal records, and unmonitored regional payroll disparities. Specifically, organizations struggle with:

- **Compensation Disconnect:** Lack of clear alignment between employee experience, age, and compensation, leading to wage compression or arbitrary pay scales.
- **Evaluation Tracking Gaps:** Incomplete annual performance evaluations, leaving management without clear data to distribute merit-based bonuses.
- **Geographic Cost Variances:** Unmonitored regional payroll variances across corporate branches (Chicago, Los Angeles, New York), creating budget overruns without proportional output.
- **Attrition & Workforce Inactivity:** High proportions of inactive employee accounts (nearly 50%), complicating headcount planning and resource optimization.

This project solves these operational bottlenecks by providing an interactive, structured Pivot-based analytical model and dashboard.

3. Tools & Technologies Used

- **Excel Formulas & Calculation Engines:** COUNTIF, AVERAGEIFS, SUMIFS, MAX/MIN, XLOOKUP, dynamic matrix formulas, and conditional logical flags.
- **Data Modeling & Aggregation:** Dynamic Pivot Tables, Pivot Charts, Timeline Slicers, and multi-variable cross-tabulations.
- **Data Transformation & Power Query:** Automated cleaning, standardized text formats, date formatting, missing value imputation, and categorical indexing.

- **Dashboard Design & UI/UX:** Custom KPI scorecards, visual hierarchy formatting, color-coded status badges, and interactive control slicers.

4. Dataset Description & Architecture

The underlying analytical dataset comprises 2,000 individual employee records spanning 12 transactional attributes:

- **Demographics:** Employee ID, Name, Age (Range: 18–65 years, Average: 41.25 years), Gender breakdown (Female: 663, Male: 653, Other: 684).
- **Workforce Operations:** Department (HR: 659, IT: 696, Sales: 645), Branch Location (Chicago: 677, Los Angeles: 678, New York: 645), Work Session (Morning: 678, Evening: 652, Night: 670), Employment Status (Active: 1,007, Inactive: 993).
- **Financials & Performance Metrics:** Monthly Base Salary (Min: \$2,015.00, Max: \$9,993.00, Mean: \$5,971.31), Work Experience (Min: 1 year, Max: 20 years, Mean: 10.27 years), Performance Rating Score (Scale 1.0–5.0, Recorded: 989, Null: 1,011).

5. Key Analytical Findings & Pivot Summaries

The core analysis leverages dynamic cross-tabulations to evaluate regional and departmental workforce metrics derived directly from the pivot dashboard:

A. Regional Experience vs. Compensation Analysis

Analysis of work experience against monthly compensation shows balanced staffing across cities, with Los Angeles maintaining the highest overall average monthly compensation:

Location	Department	Average Experience (Years)	Average Salary (\$)
Chicago	HR	10.05	5,919.81
	IT	10.60	6,084.44
	Sales	9.33	5,847.54
	Subtotal	10.00	5,951.57
Los Angeles	HR	10.96	6,417.48
	IT	9.89	5,959.52
	Sales	10.43	5,847.50
	Subtotal	10.41	6,068.58
New York	HR	11.08	5,864.23
	IT	9.99	5,758.29
	Sales	10.21	6,062.57
	Subtotal	10.41	5,889.78
Grand Total	All Departments	10.27	5,971.31

Table 1: Cross-Tabulation of Experience and Salary by Location and Department

B. Employment Status & Active Headcount Distribution

The active workforce ratio remains close to 50% across all geographic centers, indicating equal attrition or inactive record retention across operating hubs:

Location	Active Employees	Inactive Employees	Total Headcount
Chicago	340	337	677
Los Angeles	341	337	678
New York	326	319	645
Grand Total	1,007	993	2,000

Table 2: Employee Status Breakdown Across Corporate Locations

Location	Department	Average Age (Years)	Average Salary (\$)
Chicago	HR	41.03	5,919.81
	IT	39.34	6,084.44
	Sales	41.47	5,847.54
	Subtotal	40.61	5,951.57
Los Angeles	HR	41.59	6,417.48
	IT	40.31	5,959.52
	Sales	42.03	5,847.50
	Subtotal	41.28	6,068.58
New York	HR	40.46	5,864.23
	IT	42.94	5,758.29
	Sales	42.17	6,062.57
	Subtotal	41.90	5,889.78
Grand Total	All Departments	41.25	5,971.31

Table 3: Demographic Breakdown: Age and Compensation Metrics

C. Demographics Analysis: Age Demographics vs. Compensation

Evaluating age alongside compensation reveals that salary allocation is distributed evenly across age bands, with average employee ages hovering closely around 40 to 42 years across all department-location pairs:

6. Deep-Dive Analytical Insights

1. **Highest Earning Segments:** Los Angeles HR professionals command the highest average salary at \$6,417.48 with an average experience of 10.96 years. Conversely, New York IT personnel represent the lowest average salary segment at \$5,758.29.
2. **Flat Wage-to-Experience Curve:** Across the organization, work experience averages 10.27 years while compensation averages \$5,971.31. The data shows minimal pay variance between junior and senior personnel, pointing to a potential area for compensation structure realignment.
3. **Performance Rating Gap:** Over 50% of active employee profiles lack an assigned Performance Score (1,011 unassigned entries), indicating that performance appraisal systems need to be systematically integrated into payroll workflows.
4. **Shift & Work Session Balance:** Headcount is distributed evenly across Morning (678), Evening (652), and Night (670) shifts, demonstrating balanced operational coverage across all hours.

7. Dashboard Architecture & UI Design

The dynamic dashboard tab incorporates key UI elements to enhance readability and decision-making:

- **Top Executive KPI Scorecards:** Summary tiles displaying Total Payroll Expense (\$11.94M), Total Headcount (2,000), Average Monthly Salary (\$5,971.31), and Average Tenure (10.27 years).
- **Interactive Slicers:** Connected slicers allowing real-time filtering by Location (Chicago, Los Angeles, New York), Department (HR, IT, Sales), and Status (Active, Inactive).

- **Conditional Formatting Alerts:** Highlighted cells flagging compensation outliers, low appraisal scores, and uncompleted performance evaluations.

8. Strategic Business Recommendations

Based on the data analysis, the following strategic steps are recommended for HR and Management:

1. **Implement Merit-Based Pay Bands:** Standardize compensation tiers based on experience and performance scores to ensure equitable salary growth and boost employee retention.
2. **Mandate Performance Appraisal Tracking:** Establish quarterly appraisal deadlines to resolve the current 50.55% gap in performance rating data.
3. **Optimize Regional HR Allocation:** Review compensation in Los Angeles HR (\$6,417.48 average) and realign salary benchmarks across regional offices.
4. **Audit Inactive Employee Accounts:** Conduct a thorough audit of the 993 inactive profiles to keep system records up to date and reduce administrative overhead.

9. Learning Outcomes

Through completing this analytics project, several advanced technical competencies were gained:

- Mastered multi-variable dynamic Pivot Table creation, custom data grouping, and cross-tabulation.
- Developed proficiency in structuring raw Kaggle HR data using Power Query and advanced cleaning routines.
- Designed an interactive executive dashboard with KPI scorecards, conditional logic alerts, and slicers.
- Derived actionable business recommendations by analyzing workforce demographics, salary distributions, and performance ratings.

10. Project Conclusion

The Employee Performance & Payroll Analytics Dashboard transforms raw workforce transactional data into a powerful decision-making system. By addressing key operational gaps—such as unrecorded performance ratings, flat pay-to-experience progression, and regional cost differences—organization leadership can optimize payroll allocation, improve employee performance evaluation, and foster sustained retention.

11. GitHub Repository Details

GitHub Username	Somnaths13, adarpitadhar13-DA
GitHub Repo Link	Somnath's Repo / Arpita's Repo